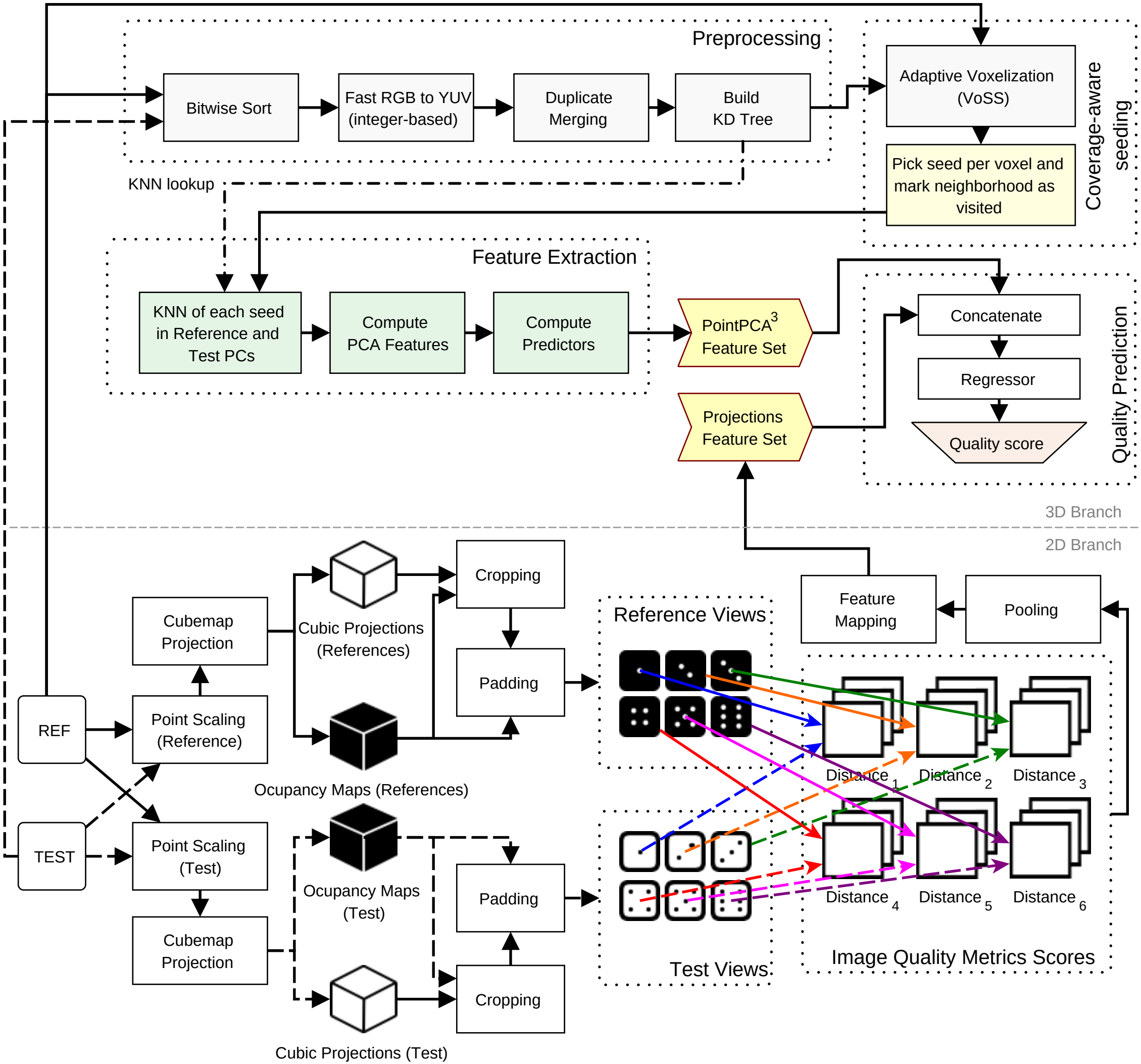
Preprocessing
Bitwise Sort
Fast RGB to YUV (integer-based)
Duplicate Merging
Build KD Tree
Coverage-aware seeding
Adaptive Voxelization (VoSS)
Pick seed per voxel and mark neighborhood as visited
KNN lookup
Feature Extraction
KNN of each seed in Reference and Test PCs
Compute PCA Features
Compute Predictors
PointPCA3 Feature Set
Projections Feature Set
Quality Prediction
Concatenate
Regressor
Quality score
3D Branch
2D Branch
Cubemap Projection
Cubic Projections (References)
Cropping
Padding
REF
Point Scaling (Reference)
Ocupancy Maps (References)
TEST
Point Scaling (Test)
Cubemap Projection
Ocupancy Maps (Test)
Padding
Cropping
Cubic Projections (Test)
Reference Views
Test Views
Feature Mapping
Pooling
Distance 1
Distance 2
Distance 3
Distance 4
Distance 5
Distance 6
Image Quality Metrics Scores